

ASSIGNMENT-2

EX 1

Classify the following continuous-time signals as even or odd signals or neither even nor odd. Determine power and energy for each incase of power or energy signal.

- (i) $x(t) = (1 + t^2)\cos^2(5t)$
- (ii) $x(t) = u(t)$
- (iii) $x(t) = tu(t)$
- (iv) $x(t) = t\sin(2t)$
- (v) $x(t) = t + \cos(2t)$
- (vi) $x(t) = e^{-2t}\sin(2t)$

EX 2

Consider the following continuous-time signal:

$$x(t) = 2 \sin \left(\frac{2\pi(t - T)}{10} \right)$$

Determine the values of T for which the signal is

- (i) An even function
- (ii) An odd function

EX 3

Show that the energy of $x(t) \pm y(t)$ is $E_x + E_y$, if $x(t)$ and $y(t)$ are orthogonal.

EX 4

Let $x_1(t)$ and $x_2(t)$ be periodic signals with fundamental periods T_1 and T_2 , respectively. Under what conditions is the sum $x(t) = x_1(t) + x_2(t)$ periodic, and what is the fundamental period of $x(t)$ if it is periodic?

EX 5

Show that the k^{th} and m^{th} harmonic of the signal $x(t) = e^{j\omega_0 t}$ are orthogonal.

EX 6

Show that the k^{th} and m^{th} harmonic of the discrete time signal $x[n] = e^{j\frac{2\pi}{M}n}$ are orthogonal.

EX 7

Show that if $x(t + T) = x(t)$, then

$$\int_{\alpha}^{\beta} x(t) dt = \int_{\alpha+T}^{\beta+T} x(t) dt$$

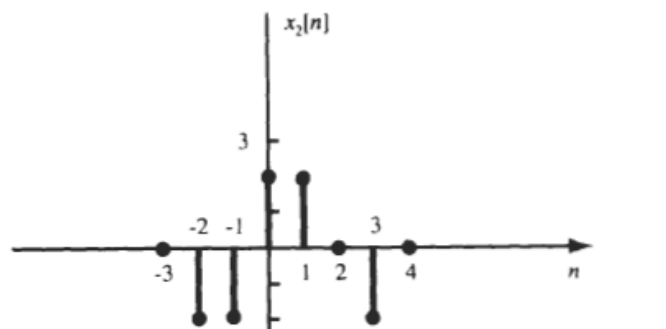
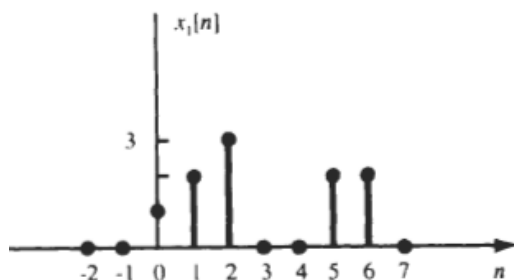
$$\int_0^T x(t) dt = \int_a^{a+T} x(t) dt$$

for any real α , β , and a .

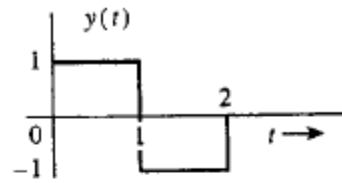
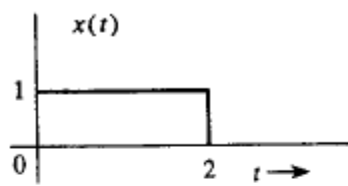
EX 8.

Using the discrete-time signals $x_1[n]$ and $x_2[n]$ shown in Fig. 1-22, represent each of the following signals by a graph and by a sequence of numbers.

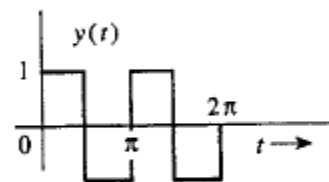
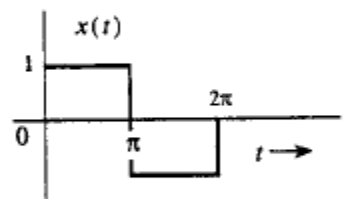
(a) $y_1[n] = x_1[n] + x_2[n]$; (b) $y_2[n] = 2x_1[n]$; (c) $y_3[n] = x_1[n]x_2[n]$



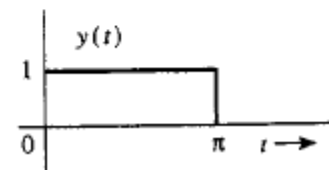
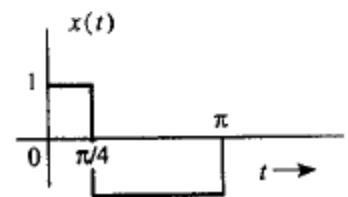
EX 9. Are $x(t)$ and $y(t)$ orthogonal to each other



(a)



(b)



(c)